

Fraser M^cEwan

St Andrews, Scotland, UK | (+44) 7960 214870

frasermcewan6@gmail.com | github.com/fmcewan | [LinkedIn](#) | fmcewan.dev

PROFILE

MSc Data-Intensive Analysis student with a BSc in Computer Science and Mathematics. Experienced in using Python to build reproducible analytical pipelines, evaluate machine-learning models and test statistical predictions across different datasets. Strong quantitative problem-solving and programming skills, with an interest in collaborating on data-driven solutions to real-world challenges.

EDUCATION

University of St Andrews Sep 2026 – Sep 2027
MSc Data-Intensive Analysis

University of Manchester Sep 2021 – Jul 2025
BSc (Hons) Computer Science and Mathematics 2:1 (66.2%)
Dissertation: Quantum Conformal Prediction *First Class (73%)*

TECHNICAL SKILLS

- **Programming:** Python, SQL, C++, C, Go, JavaScript
- **Data & Machine Learning:** NumPy, pandas, SciPy, scikit-learn, PyTorch, TensorFlow, Keras
- **Analytical Methods:** Statistical analysis, model evaluation, experimental design, conformal prediction, Bayesian optimisation
- **Tools:** Git, Linux, Docker, SQLite, PostgreSQL, GitHub Actions

PROJECTS

Quantum Conformal Prediction Aug 2024 – Jul 2025
Python, PyTorch, NumPy, pandas, Qiskit, SQLite [GitHub](#)

- Built a configurable research pipeline to train parameterised quantum models, collect measurement data, store results in SQLite and calibrate predictions for regression, classification and unsupervised learning.
- Tested models across multiple synthetic data distributions and scoring methods, using reproducible experiments to compare empirical coverage and prediction-set informativeness.
- Produced plots and quantitative metrics to assess how model choices and calibration affected statistical performance.

RGB-D Object Classification Feb 2025 – Apr 2025
Python, TensorFlow, Keras, Keras Tuner, NumPy [GitHub](#)

- Prepared colour and depth images as normalised four-channel inputs and trained an early-fusion neural network to classify objects across 51 categories.
- Used Bayesian hyperparameter optimisation to tune network architecture, dropout and learning rate, then compared validation results across configurations.
- Analysed training curves and confusion matrices to identify overfitting and prioritise model improvements.

Chess Engine Jul 2023 – Present
C++, SDL2, GitHub Actions [GitHub](#)

- Built a C++ decision-making engine with minimax search, alpha-beta pruning, move ordering and positional evaluation.
- Verified move generation and game-state behaviour with unit tests, perft validation and continuous integration.

WORK EXPERIENCE

The Horseshoe Inn – Kitchen Porter Nov 2022 – Sep 2024
Eddleston, Scotland

- Collaborated with colleagues during busy services, prioritising tasks and maintaining reliable performance under pressure.
- Trained new staff on procedures and daily workflows while maintaining cleanliness and food-safety standards.